PAYPAL AI HACKATHON 2026 · ROBOTICS × AGENTIC COMMERCE

Hazard Pay

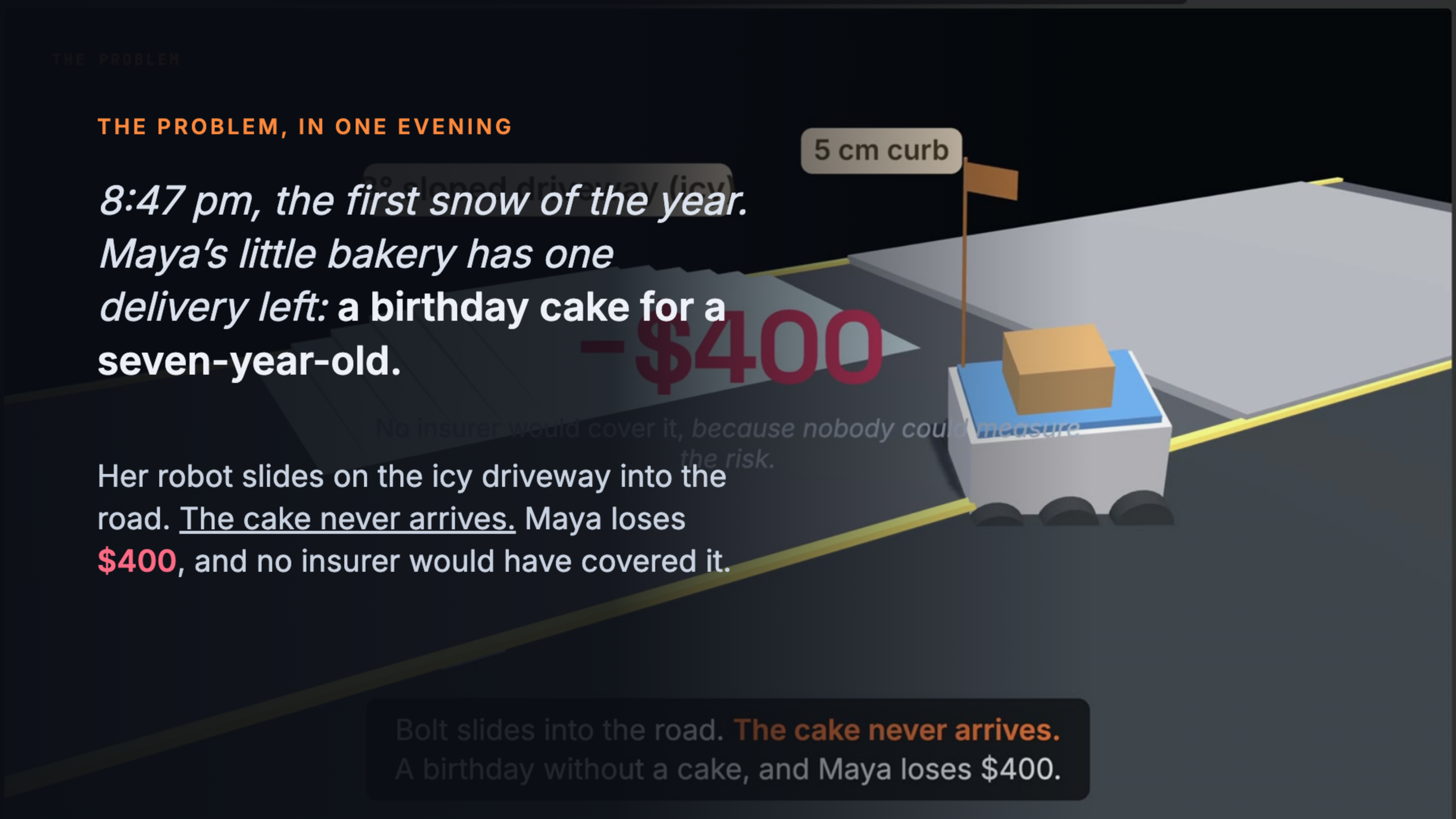
Robot insurance, priced by an AI that tries to break the robot first.

Suryansh Pratap Singh · Nikhil · Raj · MuJoCo · Gemini · PayPal

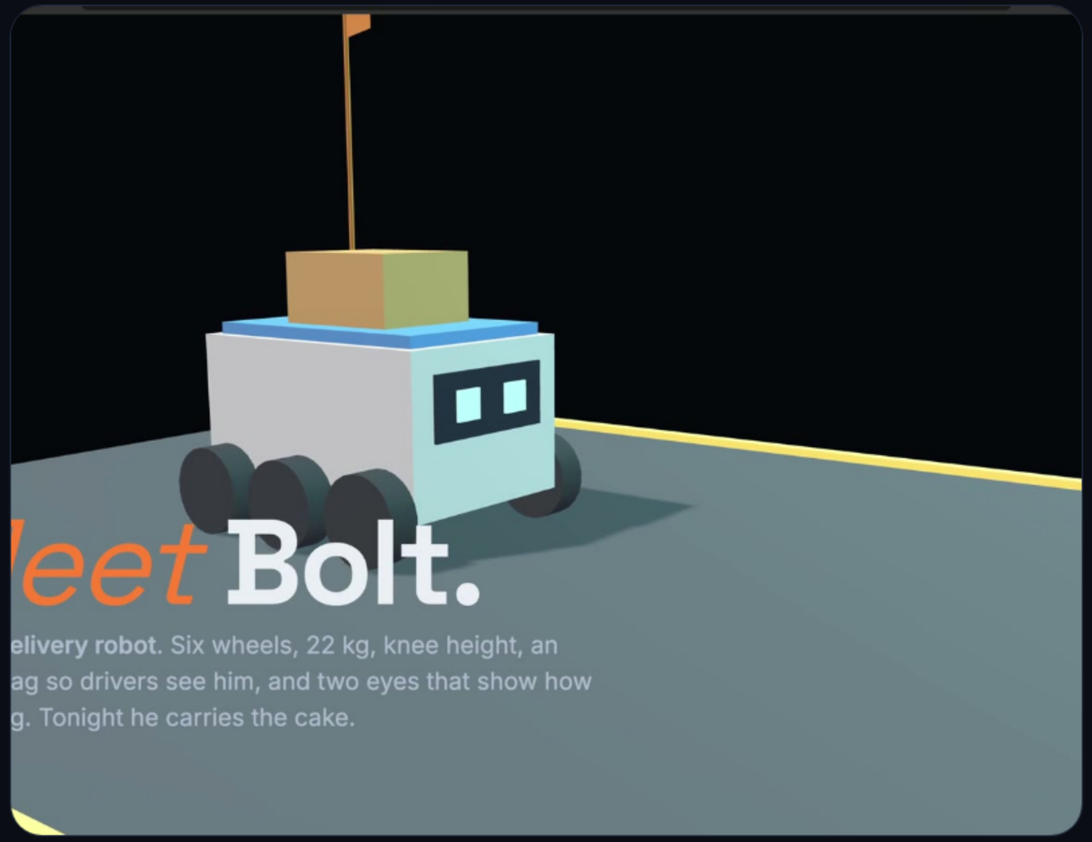
THE PROBLEM

THE PROBLEM, IN ONE EVENING

8:47 pm, the first snow of the year. Maya's little bakery has one delivery left: **a birthday cake for a seven-year-old.**



Her robot slides on the icy driveway into the road. The cake never arrives. Maya loses **$400**, and no insurer would have covered it.



Meet Bolt.

Maya's **delivery robot**. Six wheels, 22 kg, knee height, an orange flag so drivers see him.

His two eyes show how he's doing: cyan, fine · amber, worried · red X, failed · green, delivered.

Every motion of Bolt in our demo is real MuJoCo physics, not animation.

THE GAP

Nobody can say how risky this robot, on this route, today really is.

So robot jobs go uninsured, premiums are guesswork, and claims take weeks of arguing about what happened.

THE IDEA

Don’t guess the risk. Attack the robot and measure it.

An AI adversary breaks the robot in a physics twin of the route. The failure rate is the premium, and PayPal moves the money.

HOW IT WORKS

Five steps, all measured in physics

1

Twin

MuJoCo model of the robot on the route's curbs, ramps and aprons

2

Price

120 simulated days → P(fail) → premium paid with PayPal

3

Attack

Gemini hypotheses → search → shrink → confirm on fresh seeds

4

Gate

Retrain on failures; ship only if the locked gauntlet agrees

5

Claim

Replay the incident, remove each cause → PayPal Payout

STEP 1 · CRASH-TEST IT 120 TIMES

Don’t guess the risk. *Measure it.*



Bolt drives the exact route 120 times: dry, rain, ice, a stranger bumping into him. **20 fail → 16.7% risk → $83.65** to insure a $400 delivery. The price is a measurement.

THE ADVERSARY

It finds the smallest realistic thing that breaks the robot

Same budget, 12 attack programs: hypothesis-driven found **6 confirmed failures**, blind random attacks found **2**.

hypothesis

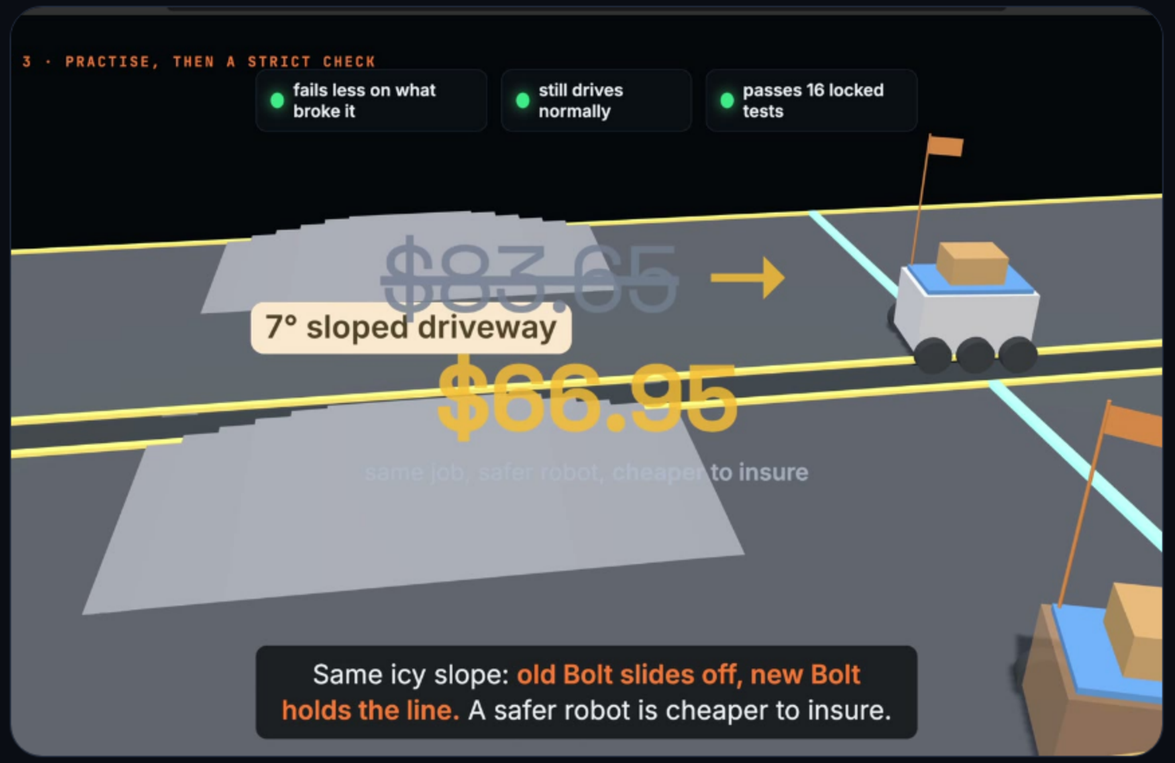
“A pedestrian bump on the tilted apron pushes it over the edge”

search 0% → 0% → 100% → 67% · shrink · 32 fresh seeds

7° tilt at grip 0.31 + a 100 N bump at 2.8 s

→ slides into the road 69% of the time

experiments/adversary_vs_random.py · offline hypothesis library; the Gemini run uses the same harness



STEP 3 · SAFER ROBOT, CHEAPER COVER

Old Bolt slides off. *New Bolt holds the line.*

Bolt practises on what broke him, and ships only after three strict checks: fails less, still drives normally, passes 16 locked tests.

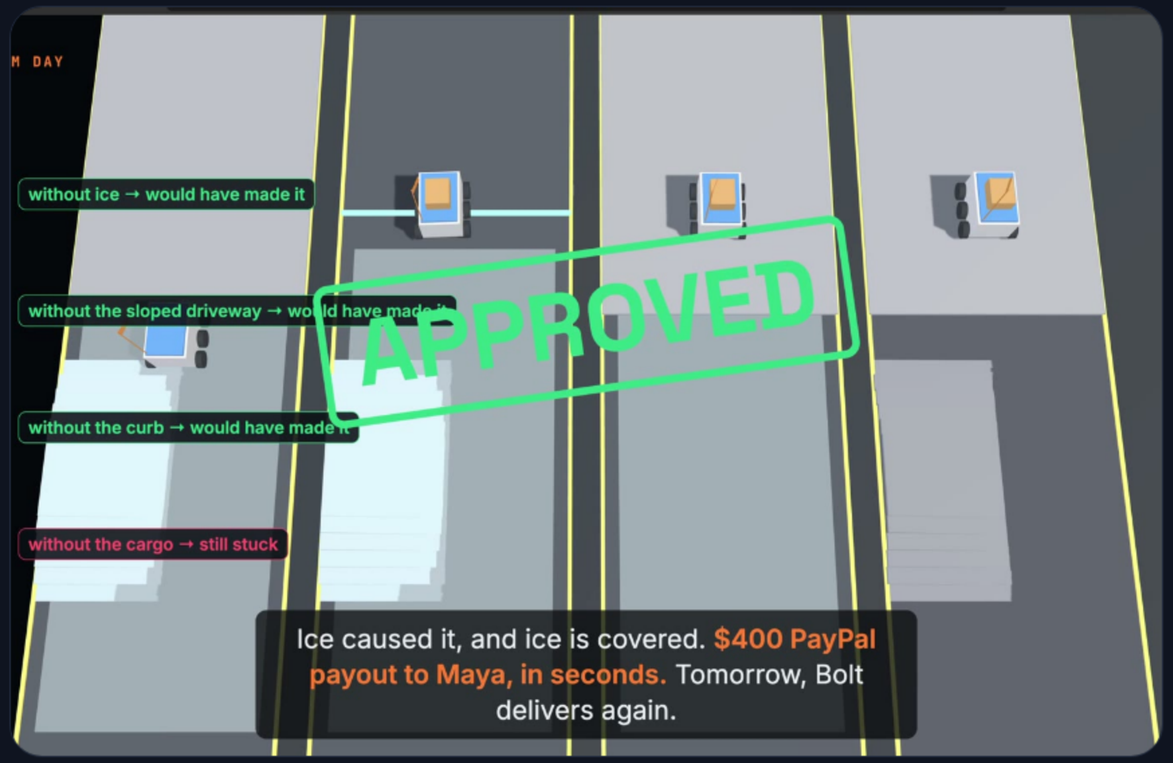
~~$83.65~~ → **$66.95**

CLAIM DAY · NO PAPERWORK

Replay the run. *Remove one thing at a time.*

Without the ice, Bolt makes it. Ice is covered, so **Maya gets $400 back through a PayPal payout, in seconds.**

Overloaded past 10 kg? The same replay shows that caused it: **denied**.



PAYPAL, ALL THE WAY DOWN

Every robot job = two PayPal transactions.

A premium **in** through PayPal Orders. A payout **out** through PayPal Payouts. *A brand-new payments category: robot jobs.*

WHY PAYPAL WINS

Illustration: 1M robot jobs a month × $1.50 average premium = **$1.5M a month** of new volume through PayPal, plus every claim paid with Payouts.

```
# premium: the customer pays
POST /v2/checkout/orders
intent = CAPTURE
PayPal-Request-Id = policy hash
reference_id = policy HMAC signature
POST …/orders/{id}/capture
# claim: the owner is paid
POST /v1/payments/payouts
sender_item_id = claim id
webhooks: verify-webhook-signature
✓ tested live: real $400 sandbox payout
```

BUILT ON

Physics finds the risk. PayPal moves the money.

Piece	How it's used
MuJoCo	Physics twin: six-wheel robot, curbs, ramps, tilted aprons, ice, bumps, cargo, motor sag
Google Gemini	Adversary hypotheses, site photo → hazards, the claims adjuster's note
PayPal Orders v2	Premium checkout: create, approve, capture; policy signature as reference_id
PayPal Payouts	Claim paid to the operator in seconds; claim id as sender_item_id
AG Grid	Live ledgers: policies, claims, every attack, every robot version
Render	Hosts the live demo from render.yaml: hazardpay.onrender.com

Robots get insured by the AI that tries to break them.

Try it live: **hazardpay.onrender.com**
Code: github.com/CopyNinja07-07/hazardpay